

Project Handoff: AI-Driven Emergency Care Coordination (Urban ED)

Last updated: 2025-08-25 • Owner: Katharina • Intended reader: next ChatGPT session

1) Project Overview

Problem. Urban EDs struggle with fragmented coordination from EMS scene → triage → ED → specialists: - Multiple handovers → information loss risk. - Manual orchestration of diagnostics (CT/X-ray/labs/US) costs time. - Bed/ICU bottlenecks, opaque capacity. - High cognitive/admin load → burnout + safety risk.

Goal. Build a **gate-engine** and an **advice-only AI assistant** to automate/support coordination workflows (not clinical decision-making) so staff focus on care.

Workflow stages (target scope). 1. **EMS On Scene:** Ingest structured prealert (trauma flag, vitals, allergies, prior admissions) and pre-notify hospital teams. 2. **Capacity Check:** Match patient needs to real-time bed/ICU availability. 3. **ED Arrival:** Assist reassessment; pre-fill diagnostic orders for human approval. 4. **Intra-ED Coordination:** Track diagnostics, notify on results, prompt follow-ups. 5. **Specialist Handoff:** Suggest optimal receiving department; coordinate transfer.

Guardrails (non-negotiable). - No autonomous diagnosis or treatment. - Human-in-the-loop approval for all orders and transfers. - Full audit logging for all proposals and user actions. - Advice-only phrasing everywhere.

KPIs / proxy metrics. Minutes saved per patient, reduced time-to-treatment for urgent cases, fewer interruptions (staff well-being), improved reliability of information flow.

2) Technical Approach (current architecture)

- **Notebook-centric prototype** (Kaggle/Colab/local safe), single-source `CONFIG` is **frozen** to prevent silent drift.
- **Event log** (JSONL) at `CONFIG["EVENT_LOG_PATH"]` for all actions/proposals.
- **UI surfaces** (ip ywidgets / Streamlit) for:
 - Equipment tracking (CSV repo + move log + QR create/scan).
 - SOP registry (auto-pull; optional crawler).
 - ICU next-bed ETA panel (simple model + manual snapshot).
 - Phase-2 calculators UI (vitals/labs/context inputs, HL7 paste, one-liners output).
- **ML bridge** loads `ed_phase2_model_thr_patched(1).joblib` (16 features) and exposes `predict_one(row)`; **threshold ~0.9988** (calibrated). *Intentionally not used for decisions yet.*
- **Gate/critics** scaffolding present (`TinyCritics` , gate engine placeholders), advice-only signals wired.

3) What's Implemented (inputs → processing → outputs)

Inputs

- **Vitals JSON** (RR, SBP, GCS, HR, Temp, AVPU, MAP, age/sex, SaO₂).
- **Labs JSON** with series support for troponin and D-dimer; other labs: creatinine, platelets, albumin, calcium, Na/Cl/HCO₃, bilirubin, INR.
- **Context JSON** (pregnancy, suspected infection, vasopressors, PE/DVT flags, comorbidities, etc.).
- **HL7 paste** (OBX segments) → parsed into the Labs JSON (troponin, D-dimer, INR, bilirubin, albumin).
- **Operational**: equipment CSVs, moves log CSV, SOP CSV, ICU JSON snapshot.

Processing (Phase-2 calculators bundle)

- **Scores**: qSOFA, MEWS, HEART (with component dropdowns), GRACE (coarse), SOFA(min), Wells-PE, Wells-DVT, PERC, PESI, sPESI, SIRS, Sepsis-3 screen, Marburg chest pain, GBS, Child-Pugh, corrected calcium, anion gap.
- **D-dimer**: converts to µg/L FEU; **age-adjusted** threshold when age>50 and not pregnant.
- **Troponin (strict ESC 0/1h for Roche hs-cTnT)**:
 - Requires **exact 60-minute** pair (no tolerance).
 - **Rule-in** if baseline ≥ 52 ng/L **or** $\Delta \geq 5$ ng/L.
 - **Rule-out** if baseline < 12 ng/L **and** $\Delta < 3$ ng/L.
 - Otherwise **observe**; if no 60-min pair → explicit "no 60-min pair (latest gap=... min)".
- Exposed in bundle as: `labs.trop_prev`, `labs.trop_curr`, `labs.trop_delta`, `labs.trop_delta_pct`, `labs.trop_rule01h` ("rule_in"|"rule_out"|"observe"|None), `labs.trop_dt_min`.

Outputs

- `one_liners`: single-line human-readable summary (scores, D-dimer vs threshold, **hs-cTnT 0/1h decision**, etc.).
- **Structured payload**:
 - `scores` dict (all calculators)
 - `labs` dict (D-dimer, troponin details, anion gap, etc.)
 - `signals` dict for gates: `acs_rulein` / `acs_ruleout` from strict 0/1h classifier (advice-only usage).
- **Event log** entries: `phase2_bundle`, `ml_score`, `pregnancy_years`, meds import events, ICU snapshot events, etc.

4) Current Position

-  **Preflight PASS**. CONFIG frozen; event logging OK; calculators OK; ESC 0/1h troponin logic integrated; signals emitted.
-  **Model bundle loaded** (16 features) and callable; **not** used to drive care flow yet (no clear label/goal).
-  **UI working**: calculators, HL7 paste, equipment/ICU panels; SOP auto-pull optional.

-  **HL7:** using `parse_hl7_labs` (targeted). A fuller `parse_hl7_all` not yet integrated.
-  **ICU capacity:** demo snapshot/manual; no live API.
-  **SOP crawler:** present but optional; can be disabled to keep notebook slim.

5) Limitations (explicit)

- **Not for clinical use;** risk scores are simplified; thresholds follow public guidance but are demo-grade.
- **Strict 60-minute pairing** for ESC 0/1h; data outside 60 min → observe/no decision.
- **No autonomous orders/transfers;** all suggestions are advice-only.
- **ML model:** generic classifier with very high threshold; no defined clinical target → predictions are not meaningful yet.
- **Data integrations:** no production HL7 feed or bed management API; ICU capacity is manual.
- **Regulatory/privacy:** none addressed in this prototype.

6) How to Run (quick)

1. **Set flags:** `CONFIG['RUN_UI']=True`; bundle path auto-loads to `/kaggle/working/ed_phase2_model_thr_patched(1).joblib`.
2. **Run preflight** cell (checks frozen config, event log, calculators, ESC 0/1h logic).
3. **Use UI:**
4. Enter **Vitals/Labs/Context**; optionally paste **HL7** OBX lines; click compute.
5. Expect an hs-cTnT line like: `hs-cTnT 0/1h: RULE-IN ... / RULE-OUT ... / OBSERVE – no 60-min pair ...`.
6. **Tail event log** at `CONFIG['EVENT_LOG_PATH']` to inspect structured outputs.

Smoke tests (1-minute): - **Rule-out:** t0=10 ng/L, t1=12 ng/L at **60 min** → RULE-OUT. - **Rule-in:** t0=20, t1=26 at **60 min** ($\Delta=6$) → RULE-IN. - **No pair:** samples 0 and +90 min → OBSERVE with message.

7) Technical Reference (keys & contracts)

- `phase2_bundle(...)` **returns:**
- `one_liners: List[str]`
- `scores: Dict[str, int/float]` (qsofa, mews, heart, grace_coarse, sofa_min, wells_pe, wells_dvt, pesi, spesi, sirs, sepsis3, gbs, marburg, child_pugh)
- `labs: { d_dimer, d_dimer_thr, trop_prev, trop_curr, trop_delta, trop_delta_pct, trop_rule01h, trop_dt_min }`
- `signals: { acs_rulein: bool, acs_ruleout: bool }`
- **Event log** (JSONL) includes entries with `type` such as `phase2_bundle`, `ml_score`, `pregnancy_years`, `med_entry`, etc. Use tail-read to fetch latest bundle if recompute isn't desirable.

8) Next Steps (prioritized, low-bloat)

1. **Define** `gate_pos` (what counts as a “positive” for gating?) and minimal label schema → give purpose to the MLP.
 2. **Map signals** → **advice-only prompts** (small UI box):
 3. `acs_rulein` → page cardiology (advice-only), prep cath-lab checklist, confirm serial ECGs.
 4. `acs_ruleout` → de-prioritize ACS, proceed with non-ACS pathways.
 5. **Unit checks for ESC 0/1h** in preflight to prevent regressions (rule-in, rule-out, no pair cases).
 6. **Finalize event schema** (fields you want for analytics: `trop_*`, `signals.*`, calculator scores).
 7. **Capacity routing**: connect ICU ETA snapshot with a simple transfer suggestion (advice-only, human approval).
 8. **(Optional later)** richer HL7 ingestion; SOP crawler harden; pin sklearn version or rebuild bundle.
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9) What the Next ChatGPT Session Should Do

- Confirm environment + bundle path; run preflight.
 - Verify hs-cTnT smoke tests pass and events contain `trop_rule01h` and `trop_dt_min`.
 - Help formalize `gate_pos` and a tiny label pipeline; propose a minimal metric (e.g., precision@k on actionable coordination tasks).
 - Sketch advice-only mapping from `signals` to 2-3 succinct prompts in the UI (no new deps).
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10) Contact & Ethos

- Focus: AI ethics, non-linear emergence, broadening access to ML skills; deconstruct patriarchal defaults.
 - Design choices emphasize **transparency**, **auditability**, and **human authority** over recommendations.
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11) Modification Contract — “Do-No-Harm” for GPT/Editors

Purpose. Protect critical behavior while allowing incremental improvements. All changes must keep the notebook slim, auditable, and advice-only.

A) Protected Surface (MUST NOT be modified or replaced)

1. **CONFIG freeze** + `_FrozenConfig` semantics and keys (incl. `EVENT_LOG_PATH`, `MODEL_BUNDLE_PATH`).
2. `_append_event(ev: dict)` signature/semantics (UTC ISO-8601, JSONL append). No batching, no schema change.
3. `phase2_bundle(vitals, labs, context)` interface + return structure (`one_liners`, `scores`, `labs`, `signals`).

4. **Troponin logic:** `assess_troponin_delta` with **strict ESC 0/1h Roche hs-cTnT** (exact 60-min pairing; rule-in if `t0≥52` or `Δ≥5`; rule-out if `t0<12` AND `Δ<3`; else observe). Preserve keys: `trop_prev/curr/delta/delta_pct`, `trop_rule01h`, `trop_dt_min`.
5. **HL7 minimal parser:** `parse_hl7_labs` contract (Trop/D-dimer/INR/Bili/Albumin). No broad parser swaps.
6. **ICU ETA helpers:** `_load_icu_status`, `_compute_next_bed_eta` behavior.
7. **ML bridge public API:** `predict_one(row)` & `FEAT` exposure. May be unused; do not change threshold/semantics.
8. **Advice-only stance** across UI text and emitted suggestions.

If a protected item truly must change, STOP and request owner approval with a written impact note.

B) Allowed Changes (Append-Only)

- Add **new cells after GUARDS/PREFLIGHT** and after existing calculator cells.
- Add **small, dependency-free utilities** (e.g., display helpers, tiny advice box) that consume existing outputs.
- Add **unit checks / smoke tests** that read functions; no rewiring.

C) Disallowed Changes

- Replacing or refactoring protected functions/classes; renaming keys; altering event schema.
- Adding external dependencies, network calls, or heavy data flows.
- Generating creative “improv” code not grounded in current objects.
- Silencing warnings by swallowing exceptions around protected logic.

D) Required Workflow for Any Edit

1. **State your intent** (1–2 lines) and **list touched cells**.
2. **Append code only**; do not edit existing protected cells.
3. **Run PREFLIGHT** → must PASS.
4. **Run smoke tests (hs-cTnT)**: rule-in, rule-out, and no-pair cases.
5. **Tail event log**: show last `phase2_bundle` entry containing `trop_rule01h` and `trop_dt_min`.
6. **Show diff proof**: counts of **new** lines vs. **modified** lines (modified must be 0 for protected cells).

E) Proof of Compliance (paste into chat)

- `PREFLIGHT: PASS` (log excerpt)
- `SMOKE: RULE-IN / RULE-OUT / NO-PAIR` (one-liners)
- `EVENT: {... trop_rule01h, trop_dt_min, ...}`
- `DIFF: +N lines, 0 modified in protected surface`
- `SIZE: Notebook size Δ < 200 KB`

F) Anti-Hallucination & Safety

- If unsure about any detail, **say so** and propose a test; do not invent APIs/fields.
- Cite exact function names/keys you are using; no assumptions.

- Maintain advice-only phrasing; no clinical decisions.

G) Rollback & Recovery

- Keep previous run's notebook and bundle path printed in logs.
- Edits must be reversible by deleting the appended cell(s) only.

H) Minimal Change Template

Intent: *what you're adding*

Touched cells: *IDs/positions*

Proof: *paste PREFLIGHT/SMOKE/EVENT/DIFF/SIZE*